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yuktha@LAPTOP-UAVR6HQ0:~$ cd OSLab
yuktha@LAPTOP-UAVR6HQ0:~/OSLab$ mkdir Practical8
yuktha@LAPTOP-UAVR6HQ0:~/OSLab$ cd Practical8
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ gedit Practical8.c

** (gedit:497): WARNING **: 08:11:01.231: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:497): WARNING **: 08:11:01.231: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ ./Practical8
-bash: ./Practical8: No such file or directory
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ ^C
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ gcc Practical8.c -o Practical8
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ ./Practical8
Memory allocated using malloc()
Values after malloc: 10 20 30 40 50
Memory resized using realloc()
Values after realloc: 10 20 30 40 50 60 70 80 90 100
Memory allocated using calloc(): 0 0 0 0 0
Memory freed successfully
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ valgrind --leak-check=full ./Practical8
Command 'valgrind' not found, but can be installed with:
sudo snap install valgrind # version 3.26.0, or
sudo apt install valgrind # version 1:3.26.0-0ubuntu1
See 'snap info valgrind' for additional versions.
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [526 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [123 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [674 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.8 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [177 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [56.3 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [159 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.5 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.9 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [437 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [86.2 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [302 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [96.7 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [190 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [457 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [90.8 kB]
Get:21 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Fetched 3975 kB in 6s (624 kB/s)
68 packages can be upgraded. Run 'apt list --upgradable' to see them.

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68 packages can be upgraded. Run 'apt list --upgradable' to see them.

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yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ sudo apt install valgrind
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Installing:

valgrind

Installing dependencies:

gdb	libdebuginfod-common	libipt2
libbabeltrace1	libdebuginfod1t64	libsource-highlight-common
libc6-dbg	libdw1t64	libsource-highlight4t64

Suggested packages:

gdb-doc gdbserver valgrind-mpi kcachegrind

Summary:

Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 68

Download size: 27.1 MB

Space needed: 111 MB / 1023 GB available

Continue? [Y/n] Y

Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod-common all 0.194-4 [12.3 kB]

Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdw1t64 amd64 0.194-4 [285 kB]

Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libbabeltrace1 amd64 1.5.11-5build1 [165 kB]

Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod1t64 amd64 0.194-4 [20.7 kB]

Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 libipt2 amd64 2.1.2-3 [49.2 kB]

Get:6 http://archive.ubuntu.com/ubuntu resolute/main amd64 libsource-highlight-common all 3.1.9-4.3build2 [64.2 kB]

Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 libsource-highlight4t64 amd64 3.1.9-4.3build2 [273 kB]

Get:8 http://archive.ubuntu.com/ubuntu resolute/main amd64 gdb amd64 17.1-2ubuntu1 [4173 kB]

Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc6-dbg amd64 2.43-2ubuntu2.4 [6406 kB]

Get:10 http://archive.ubuntu.com/ubuntu resolute/main amd64 valgrind amd64 1:3.26.0-0ubuntu1 [15.7 MB]

Fetch 27.1 MB in 13s (2047 kB/s)

Preconfiguring packages ...

Selecting previously unselected package libdebuginfod-common.

(Reading database ... 48378 files and directories currently installed.)

Preparing to unpack .../0-libdebuginfod-common_0.194-4_all.deb ...

Unpacking libdebuginfod-common (0.194-4) ...

Selecting previously unselected package libdw1t64:amd64.

Preparing to unpack .../1-libdw1t64_0.194-4_amd64.deb ...

Unpacking libdw1t64:amd64 (0.194-4) ...

Selecting previously unselected package libbabeltrace1:amd64.

Preparing to unpack .../2-libbabeltrace1_1.5.11-5build1_amd64.deb ...

Unpacking libbabeltrace1:amd64 (1.5.11-5build1) ...

Selecting previously unselected package libdebuginfod1t64:amd64.

Preparing to unpack .../3-libdebuginfod1t64_0.194-4_amd64.deb ...

Unpacking libdebuginfod1t64:amd64 (0.194-4) ...

Selecting previously unselected package libipt2.

Preparing to unpack .../4-libipt2_2.1.2-3_amd64.deb ...

Unpacking libipt2 (2.1.2-3) ...

Selecting previously unselected package libsource-highlight-common.

Preparing to unpack .../5-libsource-highlight-common_3.1.9-4.3build2_all.deb ...

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Preparing to unpack .../6-libsource-highlight4t64_3.1.9-4.3build2_amd64.deb ...
Unpacking libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Selecting previously unselected package gdb.
Preparing to unpack .../7-gdb_17.1-2ubuntu1_amd64.deb ...
Unpacking gdb (17.1-2ubuntu1) ...
Selecting previously unselected package libc6-dbg:amd64.
Preparing to unpack .../8-libc6-dbg_2.43-2ubuntu2.4_amd64.deb ...
Unpacking libc6-dbg:amd64 (2.43-2ubuntu2.4) ...
Selecting previously unselected package valgrind.
Preparing to unpack .../9-valgrind_1%3a3.26.0-0ubuntu1_amd64.deb ...
Unpacking valgrind (1:3.26.0-0ubuntu1) ...
Setting up libdebuginfod-common (0.194-4) ...
Setting up libsource-highlight-common (3.1.9-4.3build2) ...
Setting up libc6-dbg:amd64 (2.43-2ubuntu2.4) ...
Setting up libdw1t64:amd64 (0.194-4) ...
Setting up libipt2 (2.1.2-3) ...
Setting up libbabeltrace1:amd64 (1.5.11-5build1) ...
Setting up valgrind (1:3.26.0-0ubuntu1) ...
Setting up libdebuginfod1t64:amd64 (0.194-4) ...
Setting up libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Setting up gdb (17.1-2ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.4) ...
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ valgrind --leak-check=full ./Practical8
==3119== Memcheck, a memory error detector
==3119== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==3119== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==3119== Command: ./Practical8
==3119==
Memory allocated using malloc()
Values after malloc: 10 20 30 40 50
Memory resized using realloc()
Values after realloc: 10 20 30 40 50 60 70 80 90 100
Memory allocated using calloc(): 0 0 0 0 0
Memory freed successfully
==3119==
==3119== HEAP SUMMARY:
==3119==   in use at exit: 0 bytes in 0 blocks
==3119==   total heap usage: 4 allocs, 4 frees, 1,104 bytes allocated
==3119==
==3119== All heap blocks were freed -- no leaks are possible
==3119==
==3119== For lists of detected and suppressed errors, rerun with: -s
==3119== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ gedit cow.c

** (gedit:3137): WARNING **: 08:17:45.746: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:3137): WARNING **: 08:17:45.747: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

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==3119== For lists of detected and suppressed errors, rerun with: -s
==3119== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ gedit cow.c

** (gedit:3137): WARNING **: 08:17:45.746: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:3137): WARNING **: 08:17:45.747: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ gcc cow.c -o cow
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ ./cow
Before fork: data[0] = 10
Child before modification: data[0] = 10
Child after modification: data[0] = 99
Parent after child modification: data[0] = 10
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ gedit cow.c

** (gedit:4412): WARNING **: 08:20:53.259: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:4412): WARNING **: 08:20:53.260: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ gcc cow.c -o cow
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ ./cow
Parent PID: 5559
Parent waiting...
Child PID: 5560
Child before modification...
Child after modification...
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ gedit cow.c

** (gedit:5633): WARNING **: 08:23:22.912: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:5633): WARNING **: 08:23:22.912: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ gcc cow.c -o cow
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ ./cow
Parent PID: 6873
Parent waiting...
Child PID: 6874
Child BEFORE modification
Child AFTER modification
yuktha@LAPTOP-UAVR6HQ0:~/OSLab/Practical8$ |

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yuktha@LAPTOP-UAVR6HQ0:~$ ps -o pid,ppid,rss,vsz,cmd -p 5559,5560
  PID    PPID   RSS   VSZ  CMD
yuktha@LAPTOP-UAVR6HQ0:~$ ps -o pid,ppid,rss,vsz,cmd -p 6873,6874
  PID    PPID   RSS   VSZ  CMD
 6873    337 42908 43732 ./cow
 6874    6873 41964 43732 ./cow
yuktha@LAPTOP-UAVR6HQ0:~$ |
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